IS 630 - Statistical Thinking for Data Science
August Term 2025-26

Why Are We Getting Heavier? Understanding Obesity Beyond the Numbers

G1 - Team 5

OBJECTIVES

890 Million
Adults Worldwide Living with **Obesity**

2.5 Billion
Adults aged > 18 being **Overweight**

7% Countries
Prepared to Manage Obesity Issues

Explore the relationship between obesity and various factors related to demographics, diet, and lifestyle.

Provide insights on the matter to increase awareness and advise decision making for relevant parties.

https://www.who.int/news-room/fact-sheets/detail/obesity-and-overweight

DATA SOURCE

Data source - 2019 study on obesity levels and lifestyle habits in Colombia, Peru, and Mexico (Palechor & De la Hoz Manotas). Contains 2,111 records with 17 attributes.

https://archive.ics.uci.edu/dataset/544/estimation+of+obesity+levels+based+on+eating+habits+and+physical+condition

Target Variable - *Obesity Levels (7 groups)*

Demographic variables
Age, Gender, Height, Weight, family_history_with_overweight

Dietary Habits
FAVC (high-calorie food consumption), *FCVC* (frequency of vegetable consumption), *NCP* (number of main meals per day), *CAEC* (consumption of food between meals)

Lifestyle Behaviours
SMOKE (smoking), *FAF* (frequency of physical activity), *TUE* (tech devices usage), *CALC* (alcohol consumption), *MTRANS* (mode of transportation), *CH2O* (water intake frequency), *SCC* (calorie monitoring)

ANALYTICAL QUESTIONS

Demographic

What are the demographic factors associated with obesity?

Is obesity associated with family history with obesity?

Dietary

What are the dietary factors associated with obesity?

How are eating habits related to obesity?

Lifestyle

What are the lifestyle factors associated with obesity?

Is it true that smoking and alcohol consumption are related to obesity?

DATA PREPARATION AND FEATURE ENGINEERING

1. Check Data Types

Categorical (Nominal) → e.g Mode of Transportation
Categorical (Ordinal) → e.g Alcohol Consumption
Binary → e.g Gender, Smoking
Numerical (Continuous) → e.g Height, Age, Weight, BMI
Numerical (Discrete) → e.g Vegetable Consumption → Converted to categorical ordinal type

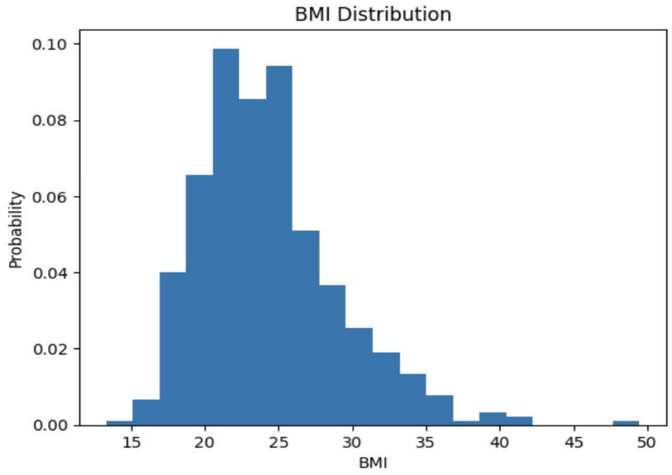
2. Remove Synthetic Data

3. Create New Variable - BMI

Underweight Less than 18.5
Normal 18.5 to 24.9
Overweight I 25.0 to 27.45
Overweight II 27.46 to 29.9
Obesity I 30.0 to 34.9
Obesity II 35.0 to 39.9
Obesity III Higher than 40

4. Checking Normality

Shapiro-Wilk Test statistic: 0.9465989254980256
p-value: 2.0569244296796905e-12



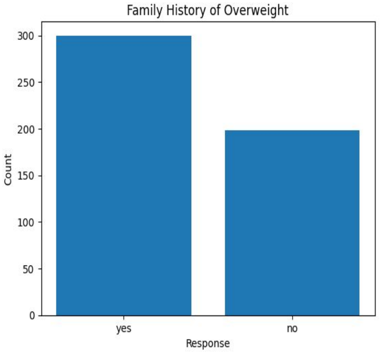
DESCRIPTIVE STATISTICS

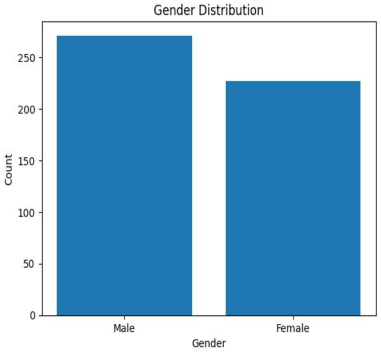
Participant Profile:

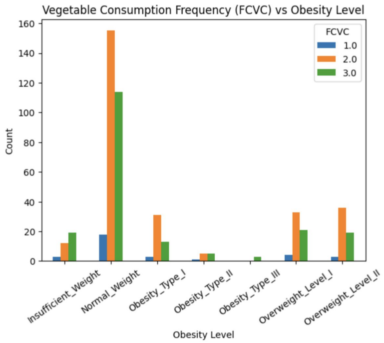
- Majority Male (271 of 498)
- Mostly young adults (mean age: 23, range; 14-61)
- Diverse weight (mean: 69.6 kg) and height (mean: 1.68m)
- Most have family history of overweight (300)

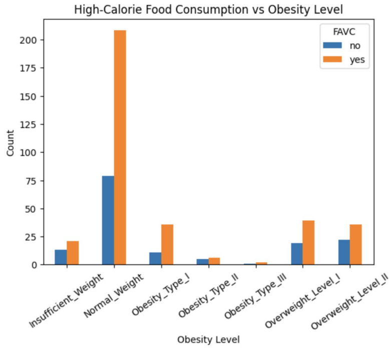
Dietary Habits:

- More than 70% frequently consume high-calorie foods (FAVC)
- Vegetable intake is low (FCVC most observed is “Sometimes”)
- Number of main meals per day is around 3
- Most eat between meals





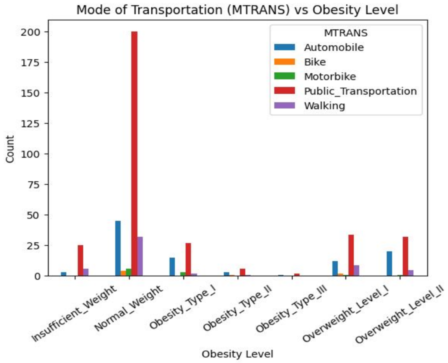


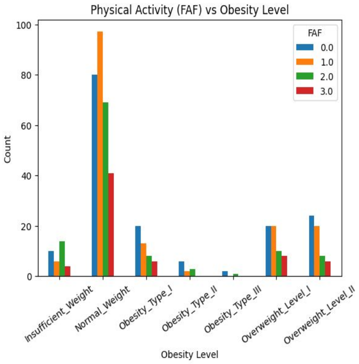


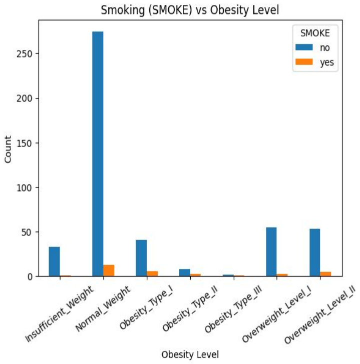
DESCRIPTIVE STATISTICS

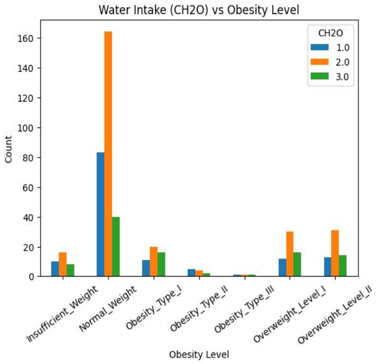
Lifestyle Behaviours:

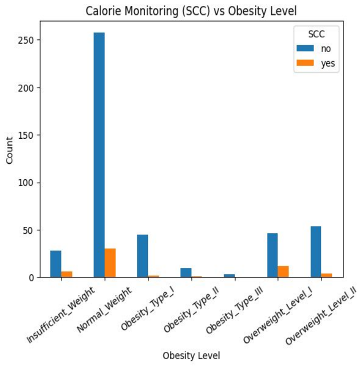
- Only 10% track calorie intake
- Physical activity is generally low
- 94% are non-smokers
- Adequate water intake (1-2L per day most common)
- Public transport is the main mode of travel











MANN-WHITNEY

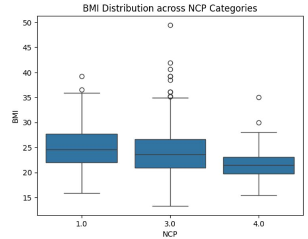
Comparing BMI distribution between 2 groups.

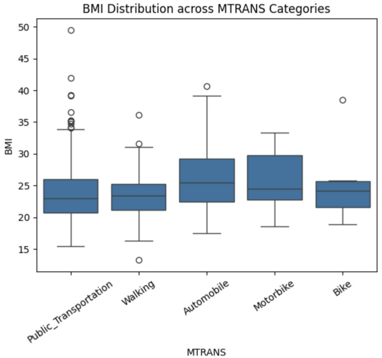
Variables	Mann-Whitney Statistics	P-value	Significance (α=0.05)
SMOKE	9723	**0.0040**	Significant
Family History with Overweight	37921	**0.00000**	Significant
Gender	25883	**0.0023**	Significant
Calorie Monitoring	11024	0.2500	Not Significant
High Calorie Consumption	25591	0.7300	Not Significant

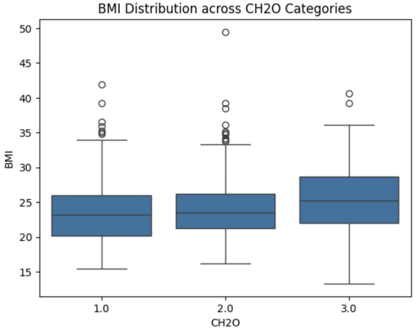
Kruskal-Wallis

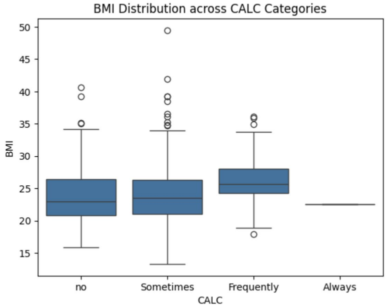
Comparing BMI distribution across groups.

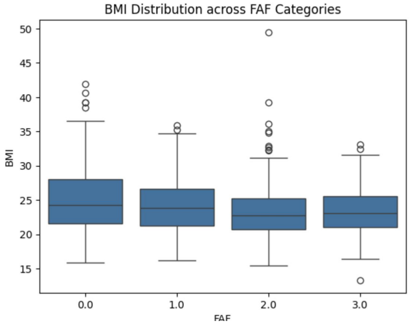
Variable	Kruskal Wallis H	P value	Significance (α=0.05)
MTRANS (Mode of Transport)	19.27	**0.0007**	Significant
CH2O (Water Intake)	10.208	**0.0070**	Significant
CALC (Alcohol Consumption)	15.39	**0.0015**	Significant
FAF (Physical Activity Frequency)	10.35	**0.0158**	Significant
TUE (Time Using Technology)	8.42	**0.0148**	Significant
CAEC (Count of Food Between Meals)	8.429	**0.0370**	Significant
FCVC (Vegetable Consumption)	1.256	0.5330	Not Significant
NCP (Main Meals)	19.29	**6.467477335470825e-05**	Significant

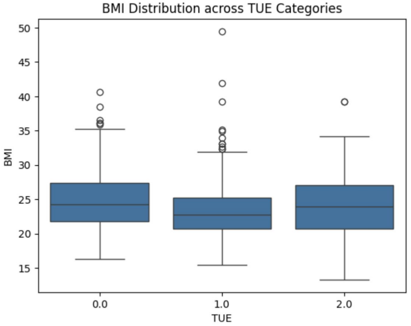


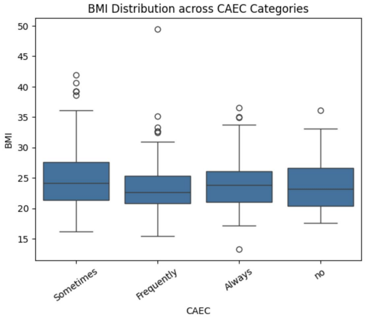




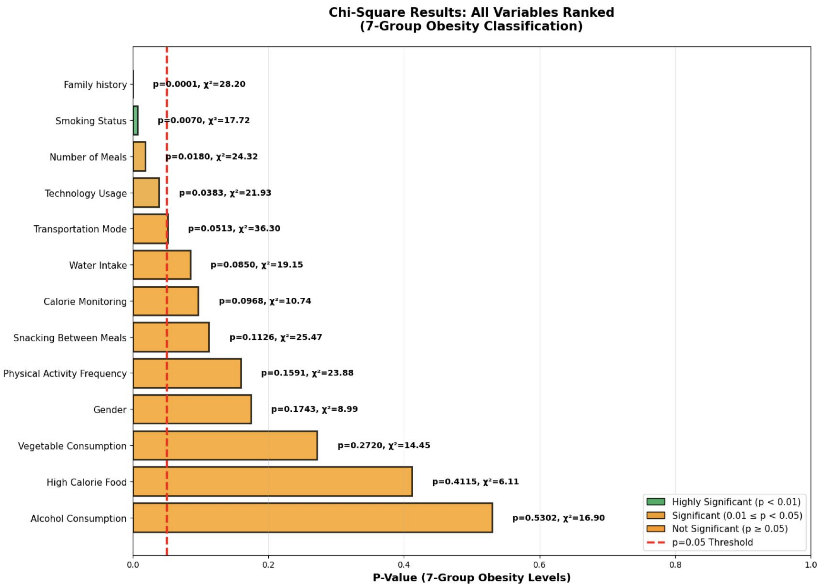








CHI SQUARE TEST



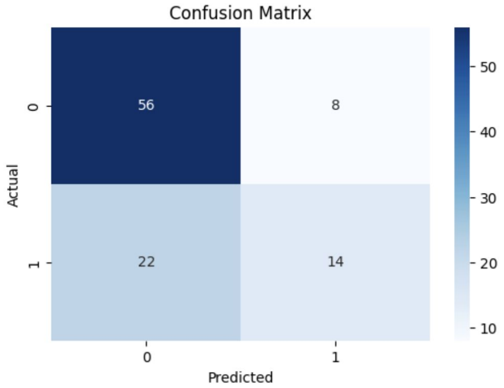
Categorical Variables	X^2 (7-Group)	P value	Significance (a=0.05)
Family History with Overweight	28.20	**0.0001**	Significant
Smoke	17.72	**0.0070**	Significant
NCP (Main Meals)	24.32	**0.0180**	Significant
TUE (Time Using Technology)	21.93	**0.0383**	Significant
MTRANS (Mode of Transport)	36.30	0.0510	Not Significant
CH2O (Water Intake)	19.15	0.0850	Not Significant
SCC (Calorie Monitoring)	10.74	0.0960	Not Significant
CALC (Alcohol Consumption)	16.90	0.5300	Not Significant
FAF (Physical Activity Frequency)	23.88	0.1591	Not Significant
FAVC (High Calorie Food Consumption)	6.11	0.4110	Not Significant
CAEC (Count of Snack Between Meals)	25.47	0.1126	Not Significant
FCVC (Vegetable Consumption)	14.45	0.2720	Not Significant
Gender	8.99	0.1740	Not Significant

LOGISTIC REGRESSION

```
Optimization terminated successfully.
         Current function value: 0.546499
         Iterations 6
                           Logit Regression Results
==============================================================================
Dep. Variable:            obese_group   No. Observations:                  498
Model:                          Logit   Df Residuals:                      481
Method:                           MLE   Df Model:                           16
Date:                Mon, 03 Nov 2025   Pseudo R-squ.:                  0.1602
Time:                        09:20:10   Log-Likelihood:                -272.16
converged:                       True   LL-Null:                       -324.07
Covariance Type:            nonrobust   LLR p-value:                 6.625e-15
==================================================================================================
                                     coef    std err          z      P>|z|      [0.025      0.975]
--------------------------------------------------------------------------------------------------
const                             -1.9770      0.893     -2.213      0.027      -3.728      -0.226
family_history_with_overweight     0.6968      0.226      3.077      0.002       0.253       1.141
NCP                               -0.3220      0.118     -2.736      0.006      -0.553      -0.091
SMOKE                              0.5794      0.433      1.337      0.181      -0.270       1.429
CH2O                               0.4271      0.165      2.583      0.010       0.103       0.751
FAF                               -0.3276      0.113     -2.902      0.004      -0.549      -0.106
TUE                               -0.0103      0.153     -0.068      0.946      -0.310       0.289
CALC                               0.1121      0.176      0.637      0.524      -0.233       0.457
Age                                0.0927      0.020      4.743      0.000       0.054       0.131
CAEC                              -0.2878      0.151     -1.900      0.057      -0.585       0.009
SCC                                0.2160      0.349      0.618      0.536      -0.469       0.901
FCVC                              -0.2696      0.189     -1.428      0.153      -0.640       0.100
Gender                             0.5922      0.227      2.609      0.009       0.147       1.037
MTRANS_Bike                       -0.1842      0.908     -0.203      0.839      -1.963       1.594
MTRANS_Motorbike                  -0.1558      0.721     -0.216      0.829      -1.568       1.257
MTRANS_Public_Transportation      -0.3863      0.299     -1.293      0.196      -0.972       0.199
MTRANS_Walking                    -0.3292      0.424     -0.777      0.437      -1.160       0.501
==================================================================================================
```

- Age (years): coef=0.0927, *p*<0.001 - Older individuals more likely to be in obese group
- Family history of overweight: coef=0.6968, p=0.002 - Family history increases likelihood of obesity risk
- NCP (Number of main meals per day): coef=–0.3220, *p*=0.006 - More main meals lower the odds of obesity (may capture healthy regular eating patterns).
- Water intake (CH2O): coef=0.4271, p=0.010 - More frequent water intake positively associated with obesity group.
- Physical activity (FAF): coef=–0.3276, p=0.004 - Higher physical activity frequency reduces obesity risk.
- Gender: coef=0.5922, p=0.009

LOGISTIC REGRESSION



Metrics	Class 0 (not obese)	Class 1 (obese)
Precision	0.718	0.636
Recall	0.875	0.389
f1-score	0.789	0.483

Accuracy = 0.700 - Model predicts correctly about 70% of the time.

Class 0 (Not obese):

- Precision = 0.718 - 71.8% of samples predicted as not obese were actually not obese.
- Recall = 0.875 - The model correctly identified 87.5% of actual not obese cases.
- F1-score = 0.789 - The model performs well in identifying the non-obese, catching most of them but with many incorrect predictions.

Class 1 (Obese):

- Precision = 0.636 - 63.6% of samples predicted as obese were actually obese.
- Recall = 0.389 - The model correctly identified 38.9% of actual obese cases.
- F1-score = 0.483 - The model misses many obese cases and generally struggles to accurately identify individuals who are obese.

INSIGHTS & FINDINGS

Category	Significant Factors	Inference
Demographic	Family History of Overweight, Gender, Age	People who are older, have a family history of overweight have a higher risk of obesity. Gender shows an association with obesity. However this finding reflects a relationship, not a definitive factor
Dietary	Number of Main Meals per Day	Individuals who have regular eating patterns tend to have lower risk of obesity
Lifestyle	Physical Activity Frequency, Technology Usage, Smoking, Water Intake	Higher physical activity is linked to lower obesity levels. Greater technology usage and smoking are related to higher obesity risk. Interestingly, higher water intake also shows a positive association with obesity.

TEAM CONTRIBUTIONS

- Akshaya - Data Preparation, Logistic Regression (Coding)
- Carina - Chi square test (Coding)
- Gyanada -Mann Whitney U test ,Kruskal Wallis test (Coding)
- Xiaoyi - Data Preparation, Descriptive statistics (Coding)
- Yuhe - Mann Whitney U test, Logistic Regression (Coding)
- Ziying - Descriptive statistics (Coding)